

Multiple Choice (10 marks)

1. Let $g(x)$ be the inverse of $f(x) = x + \cos(x)$. What is $g'(1)$?
 - (a) $\sqrt{2}$
 - (b) -2
 - (c) 1
 - (d) 0
 - (e) $-1/2$
2. Find the degree for the given field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$.
 - (a) 3
 - (b) 10
 - (c) 0
 - (d) 12
 - (e) 5
3. In a group of 10 scores, the largest score is increased by 40 points. What will happen to the mean?
 - (a) It will remain the same.
 - (b) It will increase by 30 points.
 - (c) It will increase by 10 points.
 - (d) It will decrease by 10 points.
 - (e) It will remain unchanged if the other scores also increase by 40 points.
4. What is the second number in the row of Pascal's triangle that has 43 numbers?
 - (a) 42
 - (b) 1
 - (c) 2
 - (d) 40
 - (e) 22
5. Let $P_5(x)$ be the fifth-degree Taylor polynomial approximation for $f(x)=\sin(x)$, centered at $x=0$. What is the Lagrange error of the polynomial approximation to $\sin(1)$?
 - (a) 0.02098
 - (b) 0.00198

- (c) 0.000198
- (d) 0.01998
- (e) 0.0198

True / False (10 marks)

Answer True or False

6. True or False: Rob uses approximately 4 boxes of cat food per month, as he feeds his cats 1 box every 5 days.
7. True or False: The number of paths from (0,0) to (10,10) that do not cross the diagonal $y = x$ is 16796.
8. True or False: When 640 is divided by 4, the quotient is 160 with a remainder of 0, which may suggest it fits the given condition.
9. True or False: The determinant of the matrix $A = \begin{bmatrix} 2 & 4 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix}$ is 30.
10. True or False: The expressions $5h + 3i$ and $15hi$ are equivalent for any variable values greater than zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of rounding in numerical data. Given that a coach rounded the number of runners at a track meet to 400, analyze the possible actual number of runners and justify your reasoning.
12. Discuss the structure of the symmetric group on 5 objects and analyze how to determine the largest order of a permutation within this group.

End of Paper